

PROJECT REPORT

ON

Digital Marketing Strategies & Customer Engagement in the Online Retail Industry

*Subject Area- Marketing & Business Analytics (Emerging Area in
Commerce)*

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CHAPTER 1 - INTRODUCTION

1.1 Background of the Study

The evolution of marketing practices has undergone significant transformation over the past few decades, particularly with the advancement of digital technologies and the widespread adoption of the internet. Traditional marketing methods, which primarily relied on mass communication channels such as print media, television, and radio, have gradually been supplemented and, in many cases, replaced by digital marketing strategies. This transition has enabled businesses to interact with customers in more direct, personalized, and measurable ways.

The emergence of e-commerce and online retail platforms has further accelerated this transformation. Online retail has grown rapidly due to increased internet penetration, the proliferation of smartphones, and changing consumer preferences for convenience and accessibility. Consumers today prefer digital platforms for browsing, comparing, and purchasing products, leading businesses to strengthen their presence in the online environment.

In this evolving landscape, customer engagement has become a critical factor for organizational success. Unlike traditional retail settings where interaction is limited, online retail platforms allow continuous engagement with customers through various

digital touchpoints. These interactions include website visits, mobile applications, social media platforms, and email communication. As a result, businesses are increasingly focusing on strategies that not only attract customers but also retain and engage them over time.

The availability of digital technologies has also enabled firms to collect and utilize customer-related information to better understand consumer behavior. This has led to the growing importance of data-driven decision-making in marketing. Businesses now aim to analyze customer interactions and preferences to design more effective marketing strategies. While the extent of analytical application may vary across organizations, the role of structured information and performance evaluation has become increasingly relevant in marketing practices.

Thus, the present study is situated in the context of the digital transformation of marketing and the growing significance of customer engagement in the online retail industry.

1.2 Research Gap

Existing literature provides extensive discussion on digital marketing strategies, customer engagement and online retailing. However, many studies either focus on conceptual aspects of customer engagement or examine specific digital marketing channels in isolation. Limited studies integrate marketing concepts with business

analytics to interpret customer behavior using publicly available datasets and visual analytical tools. Moreover, there is relatively less emphasis on demonstrating how descriptive analytics and interactive dashboards can support managerial decision-making in the online retail industry. The present study attempts to bridge this gap by combining insights from existing literature with an exploratory analysis of customer behavior data using business analytics techniques.

1.3 Digital Marketing in Online Retail

Digital marketing refers to the use of digital channels and technologies to promote products and services, communicate with customers, and facilitate transactions. In the context of online retail, digital marketing plays a central role in influencing consumer behavior and driving business performance.

Online retailers make use of a wide range of digital marketing channels to reach their target audience. These include search engine marketing, social media platforms, email marketing, display advertising, content marketing, and influencer collaborations. Each of these channels serves a specific purpose in the customer journey, from creating awareness to encouraging purchase and post-purchase engagement.

Search engine marketing enables businesses to increase their visibility when customers search for products or services online. Social media platforms provide

opportunities for interaction, brand building, and customer feedback. Email marketing is often used for personalized communication, promotional offers, and customer retention. Content marketing helps in providing valuable information to customers, thereby enhancing trust and engagement.

The effectiveness of digital marketing strategies depends on how well these channels are integrated and aligned with customer needs. Online retail platforms often adopt a multi-channel approach, ensuring that customers experience consistent communication across different platforms. This integrated approach is essential for maintaining customer interest and encouraging repeat interactions.

In addition, digital marketing allows businesses to track and measure the performance of their campaigns more effectively than traditional methods. Metrics such as website traffic, click-through rates, conversion rates, and customer retention levels provide insights into the effectiveness of marketing strategies. These insights can be used to refine marketing efforts and improve overall performance.

1.3.1 Concept of Customer Engagement

Customer engagement is a key concept in modern marketing, particularly in digital environments. It refers to the level of interaction, involvement, and connection that customers have with a brand or organization over time. Unlike traditional measures

of success that focused primarily on sales, customer engagement emphasizes ongoing relationships and long-term value.

Customer engagement can be understood as a multi-dimensional construct that includes behavioral, cognitive, and emotional aspects. Behavioral engagement refers to actions such as website visits, clicks, purchases, and participation in online activities. Cognitive engagement involves the level of attention and interest that customers show towards a brand, while emotional engagement relates to feelings such as satisfaction, trust, and loyalty.

In the online retail context, customer engagement is influenced by various factors, including the quality of user experience, relevance of content, personalization of offerings, and ease of navigation. Digital platforms provide multiple opportunities for interaction, allowing customers to explore products, read reviews, compare alternatives, and share feedback.

High levels of customer engagement are often associated with increased customer satisfaction, repeat purchases, and positive word-of-mouth. Therefore, businesses strive to design marketing strategies that enhance engagement across different stages of the customer journey.

1.3.2 Role of Data and Analytics in Marketing

The increasing availability of digital interaction data has led to a growing emphasis on the use of analytics in marketing. Businesses are now able to collect information related to customer behavior, preferences, and interactions across various digital platforms. This information can be organized and analyzed to support marketing decisions and improve performance.

Business analytics in marketing involves the use of data to evaluate the effectiveness of marketing strategies, identify patterns, and derive insights. Business analytics enables organizations to transform raw customer information into meaningful business insights through descriptive, diagnostic and visual analysis. Spreadsheet applications, database query tools and business intelligence platforms are increasingly used to summarize customer behavior, evaluate marketing effectiveness and support evidence-based decision-making. These analytical techniques complement traditional marketing research by improving the interpretation and presentation of secondary data. Analytical approaches may include the examination of trends, comparison of different marketing channels, and assessment of customer engagement indicators. These insights help managers in understanding what strategies are working and where improvements are needed.

In addition to analytics, recent developments in artificial intelligence have introduced new possibilities for personalization and automation in marketing. Applications such as recommendation systems, automated targeting, and predictive analysis are increasingly discussed in the context of online retail. These tools aim to enhance customer experience by providing relevant and timely information.

However, the application of analytics and artificial intelligence varies across organizations depending on resources, data availability, and strategic priorities. Therefore, while these technologies are important, their role in marketing is often supportive and complementary to broader strategic decision-making.

1.4 Need for the Study

The rapid growth of online retail and digital marketing has created a dynamic business environment where customer engagement plays a crucial role in determining success. Organizations are increasingly required to understand how different marketing strategies influence customer behavior and how engagement can be enhanced through effective communication and interaction.

Despite the availability of various digital marketing tools and techniques, there is a need to systematically examine how these strategies contribute to customer engagement in the online retail context. The complexity of digital platforms and the

diversity of customer interactions make it important to analyze existing practices and identify key factors influencing engagement.

Furthermore, the growing emphasis on data-driven decision-making has highlighted the relevance of business analytics in marketing. Understanding how analytical approaches can support marketing evaluation and strategy formulation is essential for both academic and practical purposes.

The present study seeks to address these aspects by examining digital marketing strategies and customer engagement through a structured review and analysis of secondary information. The study aims to provide insights that are relevant to both academic understanding and practical application in the field of commerce.

The study also demonstrates how business analytics tools can be applied to secondary customer datasets for generating meaningful managerial insights. This practical analytical perspective complements the conceptual understanding developed through the review of literature.

1.5 Scope of the Study

The scope of the study is limited to the examination of digital marketing strategies and customer engagement in the online retail industry. The study focuses on

understanding how different digital marketing channels and practices influence customer interaction and involvement.

The analysis is based primarily on secondary data sources, including published literature, industry reports, and documented case studies. The study does not involve primary data collection such as surveys or interviews.

The study also includes a conceptual understanding of the role of business analytics and artificial intelligence in marketing. However, the focus remains on interpretation and discussion rather than technical implementation.

The findings of the study are expected to provide a general understanding of digital marketing practices and customer engagement patterns in online retail. Since the study is based on secondary information and a publicly available customer dataset, the findings are exploratory in nature and intended to provide managerial insights rather than universal generalizations.

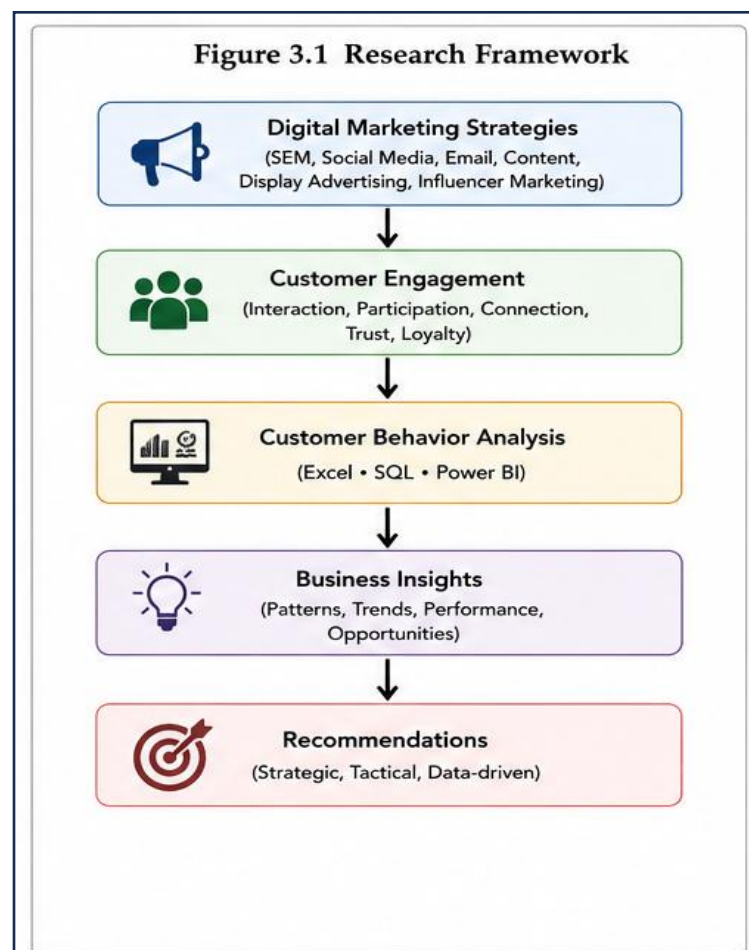
1.6 Research Questions

The present study is guided by a set of research questions that are intended to achieve the stated research objectives. The detailed research questions forming the basis of the study are presented in Chapter 3 (Research Methodology).

Figure 1.1 Conceptual Framework



Figure 1.2 Research Framework



1.7 Organization of the Study

The dissertation is organized into nine chapters. Chapter 1 introduces the background, research gap, need, scope and research questions of the study. Chapter 2 presents a review of relevant literature on digital marketing, customer engagement, business analytics and artificial intelligence in online retail. Chapter 3 explains the research methodology, including the research design, data sources, analytical tools and techniques employed. Chapter 4 presents the conceptual understanding of digital marketing and customer engagement in the online retail context. Chapter 5 presents the results and discussion based on exploratory analysis using Microsoft Excel, SQL and Power BI. Chapter 6 summarizes the major findings and conclusions of the study. Chapter 7 presents strategic and managerial recommendations. Chapter 8 discusses the academic, practical and broader implications of the study. Chapter 9 presents the limitations of the study and scope for future research.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to Literature Review

A review of literature is an essential component of any research study as it provides a foundation for understanding existing knowledge on the selected topic. It helps in identifying key concepts, theoretical perspectives, and research findings that are relevant to the study. In the context of digital marketing and customer engagement, literature review plays a significant role in understanding how marketing strategies have evolved and how customer interactions are influenced in the online environment.

The present chapter reviews existing studies and conceptual discussions related to digital marketing, customer engagement, the role of analytics, and emerging technological developments in online retail. The purpose of this review is to identify key themes, patterns, and gaps that can guide the present study.

2.1.1 Purpose of Literature Review

The literature review provides the theoretical foundation for the present study by examining previous research related to digital marketing strategies, customer

engagement, business analytics, and artificial intelligence in online retail. It helps identify established findings, emerging trends, and research gaps that justify the need for the present study.

2.2 Digital Marketing Concepts

Digital marketing has emerged as a dominant approach in modern business practices due to the increasing use of digital technologies. It involves the use of online platforms and tools to promote products and services, communicate with customers, and facilitate transactions. Unlike traditional marketing, digital marketing allows for real-time interaction and measurable outcomes.

According to Kotler et al. (2017), digital marketing integrates online and offline customer experiences, enabling businesses to create a seamless customer journey. The shift from traditional to digital marketing has been driven by changes in consumer behavior, technological advancements, and the need for personalized communication.

Kannan and Li (2017) emphasized that digital marketing has shifted organizational focus from mass communication to customer-centric engagement by leveraging digital technologies and data-driven strategies.

Chaffey and Ellis-Chadwick (2020) highlight that digital marketing strategies involve multiple channels such as search engines, social media, email, and content platforms. The effectiveness of these strategies depends on how well they are aligned with customer needs and preferences.

2.3 Customer Engagement Studies

Customer engagement has gained significant attention in recent marketing research as organizations focus on building long-term relationships with customers. It is considered a key indicator of customer loyalty and business success.

Lemon and Verhoef (2016) describe customer engagement as a multi-stage process that involves interactions across various touchpoints in the customer journey. These interactions include pre-purchase, purchase, and post-purchase activities, each contributing to the overall customer experience.

Vivek, Beatty and Morgan (2012) suggested that customer engagement extends beyond transactions and includes psychological involvement, participation and long-term relationship building between customers and brands.

Engagement is often measured through indicators such as frequency of visits, duration of interaction, repeat purchases, and customer feedback. Researchers emphasize that engaged customers are more likely to develop positive attitudes

towards a brand and contribute to its growth through continued interaction and advocacy.

2.4 Digital Channels and Consumer Behavior

The role of digital channels in influencing consumer behavior has been widely discussed in existing literature. Online platforms provide customers with access to information, reviews, and comparisons, which influence their decision-making process.

Tiago and Veríssimo (2014) explain that social media platforms have become important tools for communication and interaction between businesses and customers. These platforms enable customers to share experiences, provide feedback, and engage with brands in real time.

Different digital channels serve different purposes in the customer journey. For instance, search engines are often used for information gathering, while social media platforms are used for interaction and engagement. Email marketing is commonly used for personalized communication and customer retention.

2.5 Role of Analytics in Marketing

The increasing availability of data has led to the growing importance of analytics in marketing. Businesses now have access to large volumes of customer-related

information, which can be used to evaluate marketing performance and support decision-making.

Ryan (2016) highlights that analytics enables organizations to measure the effectiveness of marketing strategies by tracking customer interactions and campaign outcomes. Analytical approaches help in identifying trends, comparing different marketing channels, and understanding customer preferences.

Wedel and Kannan (2016) observed that advances in marketing analytics enable organizations to transform customer data into actionable insights for strategic decision-making and improved marketing effectiveness.

While analytics provides valuable insights, its effectiveness depends on how the data is interpreted and used by managers. Marketing analytics supports evidence-based decision-making by enabling organizations to evaluate customer behavior, measure campaign effectiveness and identify opportunities for strategic improvement.

2.6 Artificial Intelligence in Online Retail

Recent developments in technology have introduced the use of artificial intelligence in marketing practices. AI-enabled systems such as recommendation engines and automated targeting mechanisms are increasingly used in online retail platforms.

Verhoef et al. (2015) discuss the transition from multi-channel to omni-channel retailing, where technology plays a key role in enhancing customer experience. AI applications aim to provide personalized recommendations and improve customer satisfaction.

Davenport, Guha, Grewal and Bressgott (2020) discussed the growing adoption of artificial intelligence in marketing through applications such as personalization, predictive analytics and automated customer interactions.

However, the effectiveness of AI-based systems depends on factors such as data quality, system design, and managerial decision-making. As a result, AI is often viewed as an evolving area that requires further research and understanding.

2.7 Research Gap

Despite the extensive literature on digital marketing and customer engagement, there is a need to integrate these concepts within the context of online retail and analyze them from a structured perspective. While several studies focus on specific aspects such as social media or personalization, fewer studies provide a comprehensive understanding of how different digital marketing strategies collectively influence customer engagement.

Additionally, although the role of analytics and artificial intelligence is widely discussed, there is a need to examine these aspects in a conceptual and practical context without focusing solely on technical implementation.

The present study seeks to address these gaps by providing an integrated view of digital marketing strategies, customer engagement, and the role of analytics in the online retail industry.

Furthermore, limited studies combine conceptual understanding with exploratory business analytics using publicly available customer datasets and interactive visualization tools, creating scope for the present study.

Figure 2.1 Literature Synthesis Framework

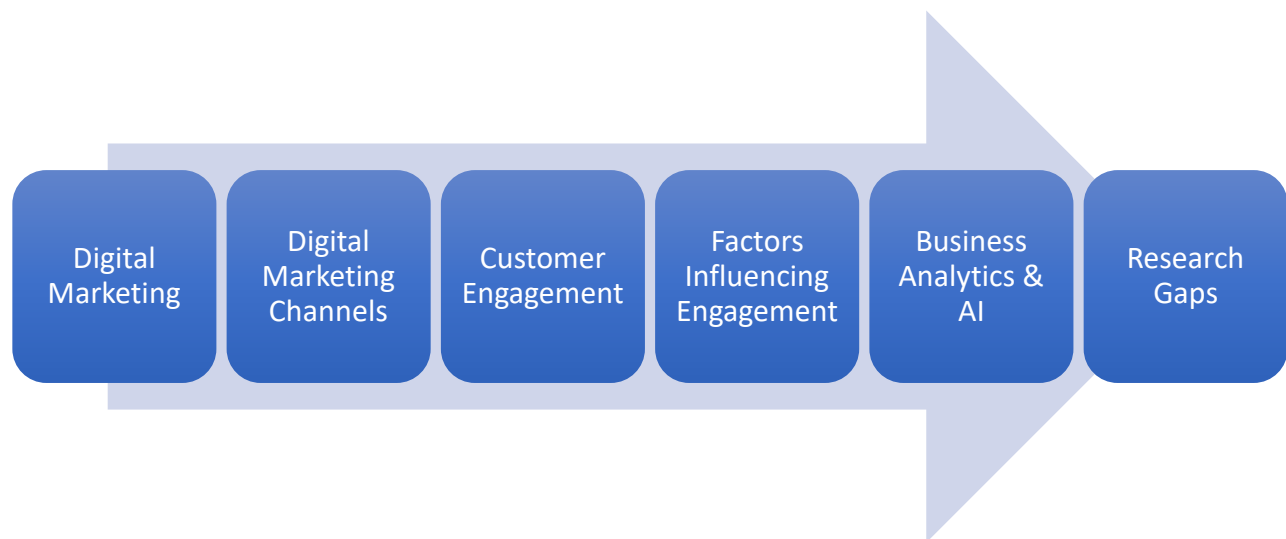


Table 2.1 Summary of Major Literature

Author	Focus Area	Major Finding
Kotler et al. (2017)	Digital Marketing	Customer-centric marketing
Chaffey & Ellis-Chadwick (2020)	Digital Channels	Multi-channel integration
Lemon & Verhoef (2016)	Customer Engagement	Customer journey
Ryan (2016)	Marketing Analytics	Data-driven decisions
Davenport et al. (2020)	Artificial Intelligence	AI in personalization

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

Research methodology refers to the systematic approach used to conduct a study and achieve its objectives. It provides a framework for collecting, organizing, and analyzing information in a structured manner. The choice of methodology depends on the nature of the research problem, availability of data, and the objectives of the study.

In the present study, the methodology is designed to examine digital marketing strategies and customer engagement in the online retail industry through a structured analysis of secondary information. The approach adopted ensures that the study remains analytical while being feasible within the scope of available data.

3.2 Rationale of the Study

The selection of the research methodology is guided by the nature of the topic and the objectives of the study. The online retail industry operates in a dynamic and data-rich environment where a large amount of information is already available through published sources, reports, and studies.

Given the availability of such information, a methodology based on secondary data is considered appropriate. This approach allows for a comprehensive understanding of digital marketing practices and customer engagement patterns without the need for primary data collection.

The methodology also supports the objective of analyzing and interpreting existing information in a structured manner, making it suitable for the present study.

3.3 Statement of the Problem

The rapid growth of digital marketing and online retail has created a complex environment where businesses must continuously adapt their strategies to engage customers effectively. Despite the availability of various digital marketing tools and platforms, understanding how these strategies influence customer engagement remains a challenge.

There is a need to examine the relationship between digital marketing strategies and customer engagement in a structured manner, taking into account the role of data and emerging technologies. The present study aims to address this problem by analyzing existing information and identifying key factors that influence customer engagement in the online retail context.

3.4 Objectives of the Study

The objectives of the study are as follows:

1. To examine digital marketing strategies adopted in the online retail industry.
2. To explore customer engagement in the context of online retail platforms.
3. To analyze key factors influencing customer engagement based on secondary information.
4. To understand the role of business analytics in marketing evaluation and decision-making.
5. To review the application of artificial intelligence-enabled techniques in online retail marketing.
6. To suggest broad recommendations for improving customer engagement strategies.

3.5 Research Questions

Research Question 1

What are the major digital marketing strategies adopted in the online retail industry?

Research Question 2

What customer engagement patterns can be observed across different customer segments in the online retail industry?

Research Question 3

What factors influence customer engagement in the online retail industry?

Research Question 4

How can business analytics support the evaluation of digital marketing performance and customer engagement?

Research Question 5

What role do artificial intelligence-enabled techniques play in enhancing customer engagement in online retail?

Research Question 6

What recommendations can enhance customer engagement strategies in the online retail industry?

3.6 Scope of the Study

The scope of the study is limited to the examination of digital marketing strategies and customer engagement in the online retail industry. The study focuses on

understanding how different marketing practices influence customer interaction and engagement.

The study is based primarily on secondary data sources and does not involve primary data collection. The analysis includes conceptual discussion as well as interpretation of available information.

The findings of the study are intended to provide a general understanding of the topic and may not be universally applicable across all industries or regions.

3.7 Research Design

The study adopts a **descriptive and analytical research design**.

The descriptive approach is used to present and explain existing digital marketing practices and customer engagement concepts. It helps in providing a clear understanding of the subject based on available information.

The analytical approach is used to interpret the collected information, identify patterns, and draw meaningful insights. This combination of descriptive and analytical design is suitable for studies based on secondary data.

3.8 Nature and Sources of Data

The study is based on **secondary data**, which is collected from various sources such as:

- academic journals and research articles
- books and published literature
- industry reports and case studies
- publicly available documents related to digital marketing and online retail

In addition to published literature and industry reports, a publicly available **E-Commerce Customer Behavior Dataset** obtained from Kaggle was used for exploratory data analysis. The dataset contains **350 customer records** and **11 variables** related to customer demographics, purchasing behavior, membership status, spending patterns and satisfaction indicators.

3.9 Data Dictionary

Table 3.1

Variable	Data Type	Description	Level	Purpose in Analysis

Customer ID	Numeric (Identifier)	Unique identifier assigned to each customer	Nominal	Used for counting customer records
Gender	Categorical	Gender of the customer	Nominal	Customer segmentation
Age	Numeric	Age of the customer in years	Ratio	Demographic analysis
City	Categorical	Customer's city	Nominal	Geographic segmentation
Membership Type	Categorical	Customer loyalty membership category (Bronze, Silver, Gold)	Ordinal	Customer engagement analysis
Total Spend	Numeric	Total amount spent by the customer	Ratio	Spending behavior analysis
Items Purchased	Numeric	Number of items purchased	Ratio	Purchasing behavior
Average Rating	Numeric	Average rating provided by the customer	Interval	Customer satisfaction indicator

Discount Applied	Categorical (Boolean)	Indicates whether a discount was applied	Nominal	Promotional strategy analysis
Days Since Last Purchase	Numeric	Number of days since the customer's last purchase	Ratio	Customer recency analysis
Satisfaction Level	Categorical	Overall customer satisfaction level	Ordinal	Primary customer engagement variable

3.10 Software and Analytical Tools Used

Table 3.2

Tool	Purpose
Microsoft Excel	Data cleaning, descriptive statistics, exploratory data analysis, PivotTables and Pivot Charts
DB Browser for SQLite	Data exploration, SQL querying, aggregation and business analysis
Microsoft Power BI	Data modelling, DAX measures, dashboard development and interactive visualization

3.11 Sampling Technique

Since the study is based on secondary data, no primary sampling is involved. Relevant sources of information are selected using a **purposive sampling approach**, where data is chosen based on its relevance to the objectives of the study.

3.12 Tools and Techniques of Data Collection

No primary data collection tools such as questionnaires or interviews are used in this study.

Data is collected through **document analysis**, which involves reviewing and analyzing published sources, reports, and literature relevant to the topic.

Secondary data were systematically organized, cleaned and validated prior to analysis to ensure consistency and reliability of the analytical results.

3.13 Tools and Techniques of Data Analysis

The collected secondary information and the publicly available customer dataset were analyzed using descriptive and exploratory analytical techniques. Microsoft Excel was used for data cleaning, descriptive statistics, PivotTables, Pivot Charts and exploratory data analysis (EDA) to summarize customer characteristics and identify behavioral patterns.

Structured Query Language (SQL), implemented using DB Browser for SQLite, was employed for data exploration, aggregation, filtering, grouping and business-oriented analytical queries. SQL analysis facilitated the examination of customer spending patterns, membership trends, satisfaction levels and inactive customer segments.

Microsoft Power BI was used to transform the dataset, create DAX measures and develop an interactive dashboard comprising KPI cards, charts and slicers. The dashboard enabled the visualization of customer behavior, spending patterns, customer satisfaction and marketing-related insights.

The analytical findings generated through these tools were interpreted to derive business insights aligned with the objectives of the study.

3.14 Limitations of Methodology

The methodology of the study has certain limitations:

- The study is based on secondary data, which may limit the depth of analysis.
- The findings depend on the availability and quality of published information.
- No primary data is collected, which may restrict direct validation of results.

However, the methodology is appropriate for achieving the objectives of the study within the given scope.

Figure 3.1 Analytical Workflow of the Study



CHAPTER 4

CONCEPTUAL AND APPLIED UNDERSTANDING

DIGITAL MARKETING AND CUSTOMER ENGAGEMENT ANALYSIS

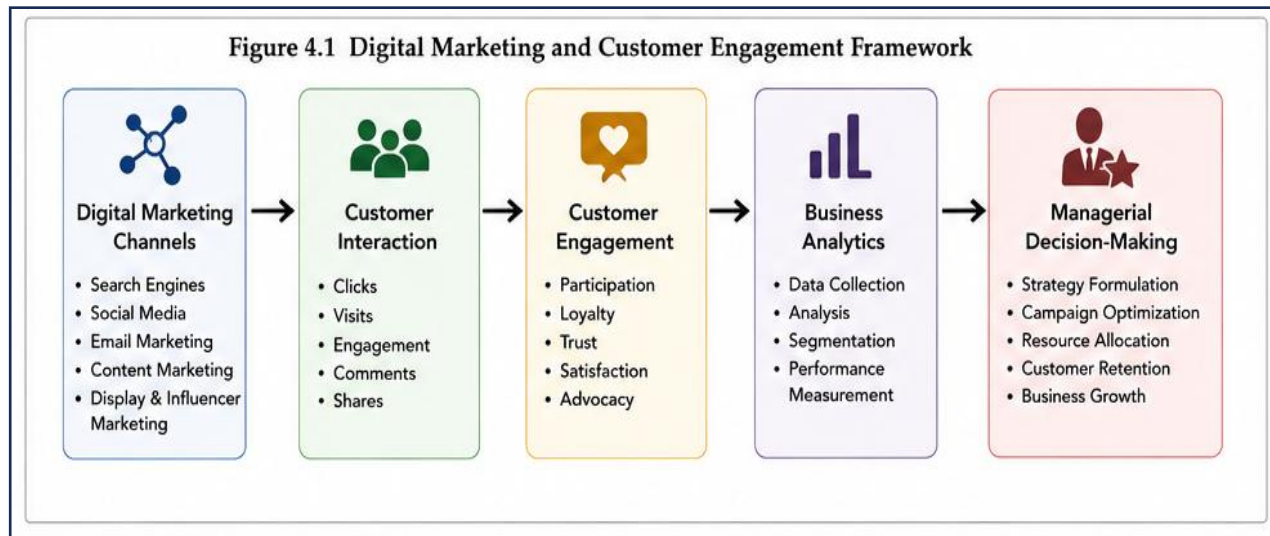
4.1 Overview of the Online Retail Industry

The online retail industry has experienced significant growth over the past decade due to advancements in technology and changing consumer behavior. The increasing use of smartphones, internet accessibility, and digital payment systems has contributed to the expansion of e-commerce platforms.

Online retail offers several advantages such as convenience, variety, competitive pricing, and ease of comparison. Consumers can browse products, read reviews, compare alternatives, and make purchases from the comfort of their homes. This shift in consumer behavior has encouraged businesses to invest in digital platforms and marketing strategies.

Major online retail platforms operate in highly competitive environments where customer acquisition and retention are critical. As a result, companies focus on developing strategies that enhance customer experience and encourage repeated interaction.

Figure 4.1 Digital Marketing and Customer Engagement Framework



4.2 Major Digital Marketing Channels

Online retailers use a variety of digital marketing channels to reach and engage customers. Each channel plays a specific role in the marketing process and contributes to overall customer engagement.

While these digital marketing channels are conceptually important, their effectiveness ultimately depends on customer engagement and behavioral responses. Therefore, the present study further explores customer behavior through secondary data analysis presented in Chapter 5.

4.2.1 Search Engine Marketing (SEM)

Search engine marketing involves promoting websites through paid advertisements and search engine optimization techniques. It helps businesses improve their

visibility in search engine results and attract potential customers actively searching for products or services.

4.2.2 Social Media Marketing

Social media platforms such as Facebook, Instagram, and Twitter are widely used for marketing purposes. These platforms enable businesses to interact with customers, share content, and build brand awareness. Social media also allows customers to provide feedback and engage with brands in real time.

4.2.3 Email Marketing

Email marketing is used for direct communication with customers. It includes promotional messages, newsletters, and personalized offers. This channel is effective in maintaining relationships with existing customers and encouraging repeat purchases.

4.2.4 Content Marketing

Content marketing focuses on providing valuable and relevant information to customers through blogs, videos, and articles. It helps in building trust and establishing long-term relationships with customers.

4.2.5 Display Advertising and Influencer Marketing

Display advertisements and influencer collaborations are used to increase brand visibility and reach a wider audience. Influencers play a significant role in shaping customer perceptions and influencing purchasing decisions.

4.3 Customer Engagement Metrics

Customer engagement in digital environments can be measured using various indicators. These metrics help businesses evaluate the effectiveness of their marketing strategies.

Some commonly used engagement metrics include:

- **Click-through rate (CTR):** Measures the number of clicks on a link relative to the number of views.
- **Website traffic:** Indicates the number of visitors to a website.
- **Time spent on platform:** Reflects the level of user interest and interaction.
- **Repeat visits:** Shows customer retention and loyalty.
- **Conversion rate:** Measures the percentage of visitors who complete a desired action, such as making a purchase.

These metrics provide valuable insights into customer behavior and help businesses identify areas for improvement.

In the present study, customer engagement is explored using indicators available in the selected secondary dataset, including customer satisfaction level, membership type, total spending, purchase recency and purchase behavior. These variables provide an exploratory understanding of customer engagement within the online retail context.

4.4 Factors Influencing Customer Engagement

Customer engagement is influenced by several factors that affect how customers interact with online platforms. Some of the key factors include:

4.4.1 Personalization

Personalized content and recommendations increase customer relevance and satisfaction. When customers receive tailored suggestions based on their preferences, they are more likely to engage with the platform.

4.4.2 User Experience

A user-friendly interface, easy navigation, and fast loading times contribute to a positive user experience. A well-designed platform encourages customers to spend more time and interact more frequently.

4.4.3 Content Quality

High-quality and relevant content helps in capturing customer attention and maintaining interest. Informative and engaging content can influence customer decisions and enhance engagement.

4.4.4 Trust and Security

Customers are more likely to engage with platforms that ensure data security and provide reliable services. Trust plays a significant role in building long-term relationships.

4.4.5 Frequency of Interaction

Regular communication through emails, notifications, and social media helps in maintaining customer engagement. However, excessive communication may lead to disengagement.

4.5 Role of Personalization in Digital Marketing

Personalization has become an important aspect of digital marketing strategies in online retail. It involves tailoring content, product recommendations, and communication based on individual customer preferences and behavior.

Online retailers use customer data to provide personalized experiences. For example, recommendation systems suggest products based on previous purchases or browsing

history. Personalized email campaigns offer targeted promotions to specific customer segments.

Personalization enhances customer satisfaction by making interactions more relevant and meaningful. It also increases the likelihood of conversion and repeat purchases. However, the effectiveness of personalization depends on the accuracy of data and the ability to interpret it effectively.

4.6 Challenges in Digital Marketing and Customer Engagement

Despite the advantages of digital marketing, businesses face several challenges in maintaining customer engagement:

- **Information overload:** Customers are exposed to a large volume of content, making it difficult to capture attention.
- **Changing consumer preferences:** Customer expectations evolve rapidly, requiring continuous adaptation.
- **Data privacy concerns:** Customers are increasingly concerned about how their data is used.
- **High competition:** Online retail platforms face intense competition, making differentiation difficult.

These challenges highlight the need for effective strategies and continuous evaluation of marketing practices.

Although these challenges influence the effectiveness of digital marketing strategies, organizations can better understand customer behavior by applying business analytics techniques to secondary customer data. In the present study, an exploratory analysis of a publicly available e-commerce customer behavior dataset was conducted using Microsoft Excel, SQL and Power BI to support the conceptual understanding presented in this chapter. The findings of this analysis are presented and discussed in Chapter 5.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction to the Analysis

The present chapter presents the results of the exploratory analysis conducted to examine customer behavior and engagement in the online retail industry using a publicly available secondary dataset. The analysis was carried out using Microsoft Excel, DB Browser for SQLite (SQL) and Microsoft Power BI. Excel was used for data cleaning, descriptive statistics and exploratory data analysis, SQL was used for querying and analyzing customer information, while Power BI was used to develop an interactive dashboard for visualization and interpretation. The findings obtained through these analytical tools are discussed in relation to the objectives of the study and existing literature.

5.2 Dataset Overview

The dataset contains customer demographic information, purchasing behavior, membership details, satisfaction levels and spending patterns. It provides an appropriate basis for conducting exploratory analysis to understand customer engagement in the online retail industry. Prior to analysis, the dataset was validated and organized to ensure consistency across the analytical tools used in the study.

The customer dataset used for the exploratory analysis was obtained from Kaggle (UOM190346A, n.d.) and contains customer demographic, purchasing and satisfaction-related variables relevant to online retail analytics.

RAW DATASET											
Customer ID	Gender	Age	Age Group	City	Membership Type	Total Spend	Items Purchased	Average Rating	Discount Applied	Days Since Last Purchase	Satisfaction Level
101	Female	29	26-30	New York	Gold	1120.20	14	4.6	TRUE		25 Satisfied
102	Male	34	31-35	Los Angeles	Silver	780.50	11	4.1	FALSE		18 Neutral
103	Female	43	41-45	Chicago	Bronze	510.75	9	3.4	TRUE		42 Unsatisfied
104	Male	30	26-30	San Francisco	Gold	1480.30	19	4.7	FALSE		12 Satisfied
105	Male	27	26-30	Miami	Silver	720.40	13	4	TRUE		55 Unsatisfied
106	Female	37	36-40	Houston	Bronze	440.80	8	3.1	FALSE		22 Neutral
107	Female	31	31-35	New York	Gold	1150.60	15	4.5	TRUE		28 Satisfied
108	Male	35	31-35	Los Angeles	Silver	800.90	12	4.2	FALSE		14 Neutral
109	Female	41	41-45	Chicago	Bronze	495.25	10	3.6	TRUE		40 Unsatisfied
110	Male	28	26-30	San Francisco	Gold	1520.10	21	4.8	FALSE		9 Satisfied
111	Male	32	31-35	Miami	Silver	690.30	11	3.8	TRUE		34 Unsatisfied
112	Female	36	36-40	Houston	Bronze	470.50	7	3.2	FALSE		20 Neutral
113	Female	30	26-30	New York	Gold	1200.80	16	4.3	TRUE		21 Satisfied
114	Male	33	31-35	Los Angeles	Silver	820.75	13	4.4	FALSE		15 Satisfied
115	Female	42	41-45	Chicago	Bronze	530.40	9	3.5	TRUE		38 Unsatisfied
116	Male	29	26-30	San Francisco	Gold	1360.20	18	4.9	FALSE		11 Satisfied
117	Male	26	26-30	Miami	Silver	700.60	12	3.7	TRUE		48 Unsatisfied
118	Female	38	36-40	Houston	Bronze	450.90	8	3	FALSE		25 Neutral
119	Female	32	31-35	New York	Gold	1170.30	14	4.7	TRUE		29 Satisfied
120	Male	34	31-35	Los Angeles	Silver	790.20	11	4	FALSE		16 Neutral

Figure 5.1: Dataset Preview

Table 5.1 Dataset Description

Attribute	Description
Dataset Name	E-Commerce Customer Behavior Dataset
Source	Kaggle

Industry	Online Retail
Number of Records	350
Number of Variables	11
Nature of Data	Secondary

5.3 Exploratory Data Analysis using Microsoft Excel

Microsoft Excel was used as the initial analytical tool for organizing, validating and exploring the customer dataset. The software facilitated data cleaning, descriptive statistical analysis, PivotTable generation and graphical representation of customer behavior. Exploratory Data Analysis (EDA) was carried out to understand the distribution of customer characteristics, identify behavioral patterns and summarize key variables before performing further analysis using SQL and Power BI.

5.3.1 Data Preparation

The customer dataset was imported into Microsoft Excel and examined for completeness and consistency. Data validation techniques were applied to verify the accuracy of records, identify missing values and ensure that all variables were assigned appropriate data types. The dataset consisted of **350 customer records** and **11 variables**, with no duplicate records identified during the validation process. The

cleaned dataset served as the foundation for subsequent descriptive analysis and visualization.

DATASET PROFILE					
Gender Distribution		Satisfaction Distribution		City Distribution	
Gender	Customer Count	Satisfaction Level	Customer Count	City	Customer Count
Female	175	Neutral	107	Chicago	58
Male	175	Satisfied	125	Houston	58
Grand Total	350	Unsatisfied	116	Los Angeles	59
		(blank)	2	Miami	58
		Grand Total	350	New York	59
Membership Distribution				San Francisco	58
Membership Type	Customer Count			Grand Total	350
Bronze	116	Discount Applied			
Gold	117	Discount Applied			
Silver	117	FALSE			
Grand Total	350	TRUE			
		Grand Total			
		350			

Figure 5.2 Dataset Profile

5.3.2 Descriptive Statistics

Table 5.2 Descriptive Statistics of Customer Variables

Variable	Minimum	Maximum	Mean	Median
Age	26	43	33.6	32.5

Total Spend	410.8	1520.1	845.38	775.2
Items Purchased	7	21	12.60	12
Average Rating	3.0	4.9	4.02	4.1
Days Since Last Purchase	9	63	26.6	23

Descriptive statistical analysis was conducted to obtain a preliminary understanding of the characteristics of the customer dataset. The analysis summarizes the central tendency and range of key variables, including customer age, spending behavior, purchase frequency, customer ratings and purchase recency.

The results indicate that the average customer age was **33.6 years**, with ages ranging from **26 to 43 years**, suggesting that the dataset primarily represents young and middle-aged consumers actively participating in online retail purchases. The mean customer expenditure was **845.38**, while the median expenditure was **775.2**, indicating the presence of several high-spending customers that increased the average spending value.

Customers purchased an average of **12.60 items**, with purchases ranging between **7 and 21 items**, reflecting moderate variation in purchasing behavior. The average

customer rating was **4.02 out of 5**, with a median rating of **4.1**, suggesting that overall customer satisfaction was relatively high across the dataset.

The average number of days since the last purchase was **26.6 days**, while the median was **23 days**. This indicates that although many customers had purchased recently, a smaller group of customers had considerably longer purchase gaps, increasing the average recency value. Overall, the descriptive statistics provide an overview of customer demographics and purchasing behavior, forming the basis for the subsequent exploratory and visualization analyzes.

The descriptive statistics indicate a generally satisfied customer base with moderate purchasing activity and noticeable variation in spending behavior, highlighting opportunities for customer segmentation and targeted marketing strategies.

Figure 5.3: Descriptive Statistical Analysis in Microsoft Excel

DESCRIPTIVE STATISTICS					
	Variable	Minimum	Maximum	Mean	Median
	Age	26	43	33.6	32.5
	Total Spend	410.8	1520.1	845.38	775.2
	Items Purchased	7	21	12.60	12
	Average Rating	3	4.9	4.02	4.1
	Days Since Last Purchase	9	63	26.6	23

5.3.3 Relationship Analysis

Relationship analysis was conducted to examine the association between customer characteristics and purchasing behavior using PivotTables and Pivot Charts in Microsoft Excel.

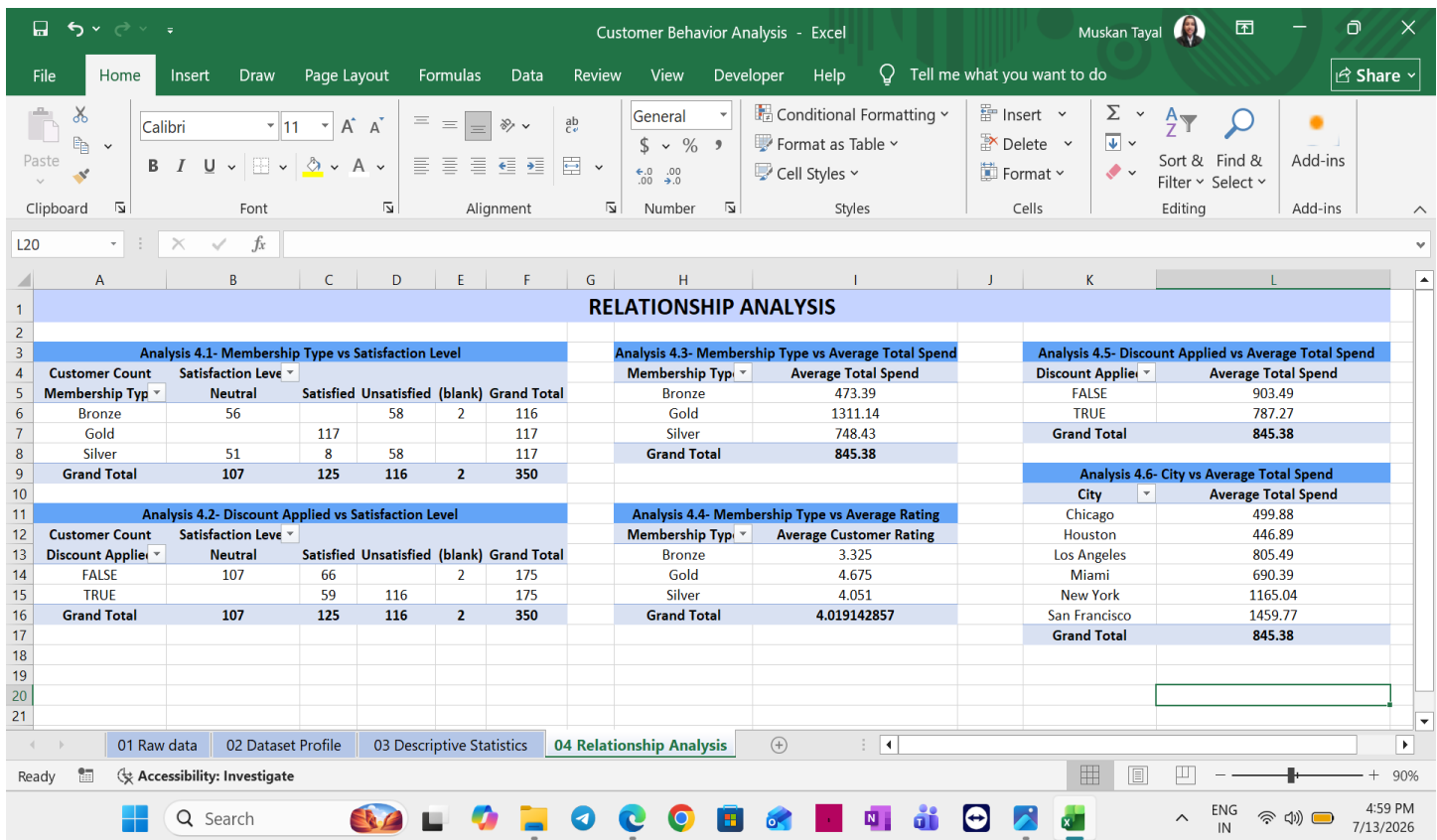


Figure 5.4: Excel Relationship Analysis

The analysis focused on customer spending, satisfaction, membership category, discount application and geographical location. These relationships provide insights

into customer engagement patterns and support the interpretation of marketing performance in the online retail industry.

5.3.3.2 Average Customer Spend by City

Customer spending was further analyzed across different cities to identify geographical variations in purchasing behavior. The analysis revealed that **San Francisco recorded the highest average customer spending (1459.77)**, followed by **New York (1165.04)**. Customers from **Houston (446.89)** and **Chicago (499.88)** recorded comparatively lower average spending levels. These variations suggest that customer purchasing behavior differs across geographical locations, highlighting opportunities for region-specific marketing strategies.

Figure 5.5 Average Customer Spend by City

Analysis 4.6- City vs Average Total Spend	
City	Average Total Spend
Chicago	499.88
Houston	446.89
Los Angeles	805.49
Miami	690.39
New York	1165.04
San Francisco	1459.77
Grand Total	845.38

5.3.3.1 Average Customer Spend by Membership Type

Membership-based spending analysis was conducted to examine the purchasing behavior of customers across different loyalty programs. The results indicate that **Gold members recorded the highest average spending (1311.14)**, followed by **Silver members (748.43)** and **Bronze members (473.39)**. The findings show that higher membership categories are associated with greater average customer expenditure. This pattern highlights the potential value of loyalty-based customer segmentation and differentiated retention strategies.

Figure 5.6 Average Customer Spend by Membership Type

Analysis 4.3- Membership Type vs Average Total Spend	
Membership Type ▾	Average Total Spend
Bronze	473.39
Gold	1311.14
Silver	748.43
Grand Total	845.38

5.3.3.3 Customer Satisfaction Distribution

The distribution of customer satisfaction levels was analyzed to understand the overall engagement of customers within the dataset. The analysis indicates that **Satisfied customers constituted the largest segment (125 customers)**, followed by **Unsatisfied customers (116)** and **Neutral customers (107)**. The relatively balanced distribution suggests that although a considerable proportion of customers reported positive experiences, a substantial number of customers remained either

neutral or dissatisfied. This highlights opportunities for online retailers to strengthen customer engagement through improved service quality, personalized marketing strategies and enhanced post-purchase experiences.

Figure 5.7 Customer Satisfaction Distribution

Analysis 4.1- Membership Type vs Satisfaction Level					
Customer Count	Satisfaction Level				
Membership Type	Neutral	Satisfied	Unsatisfied	(blank)	Grand Total
Bronze	56		58	2	116
Gold		117			117
Silver	51	8	58		117
Grand Total	107	125	116	2	350

5.3.3.4 Customer Satisfaction by Discount Status

The relationship between discount application and customer satisfaction was examined using a PivotTable. Among customers who **received discounts**, **116** were satisfied and **59** reported a neutral level of satisfaction. Among customers **who did not receive discounts**, **107** customers reported a neutral level of satisfaction while **66** were satisfied. The results indicate that customer satisfaction is not solely dependent on discount availability, suggesting that other factors such as product quality, service experience and perceived value also contribute to overall customer satisfaction.

Figure 5.8 Customer Satisfaction by Discount Status

Analysis 4.2- Discount Applied vs Satisfaction Level					
Customer Count	Satisfaction Level				
Discount Applied	Neutral	Satisfied	Unsatisfied	(blank)	Grand Total
FALSE	107	66		2	175
TRUE		59	116		175
Grand Total	107	125	116	2	350

5.3.3.5 Supporting exploratory analyses

Analysis 4.4- Membership Type vs Average Rating	
Membership Type	Average Customer Rating
Bronze	3.325
Gold	4.675
Silver	4.051
Grand Total	4.019142857

Analysis 4.5- Discount Applied vs Average Total Spend	
Discount Applied	Average Total Spend
FALSE	903.49
TRUE	787.27
Grand Total	845.38

Figure: 5.9 & 5.10 Supporting analysis

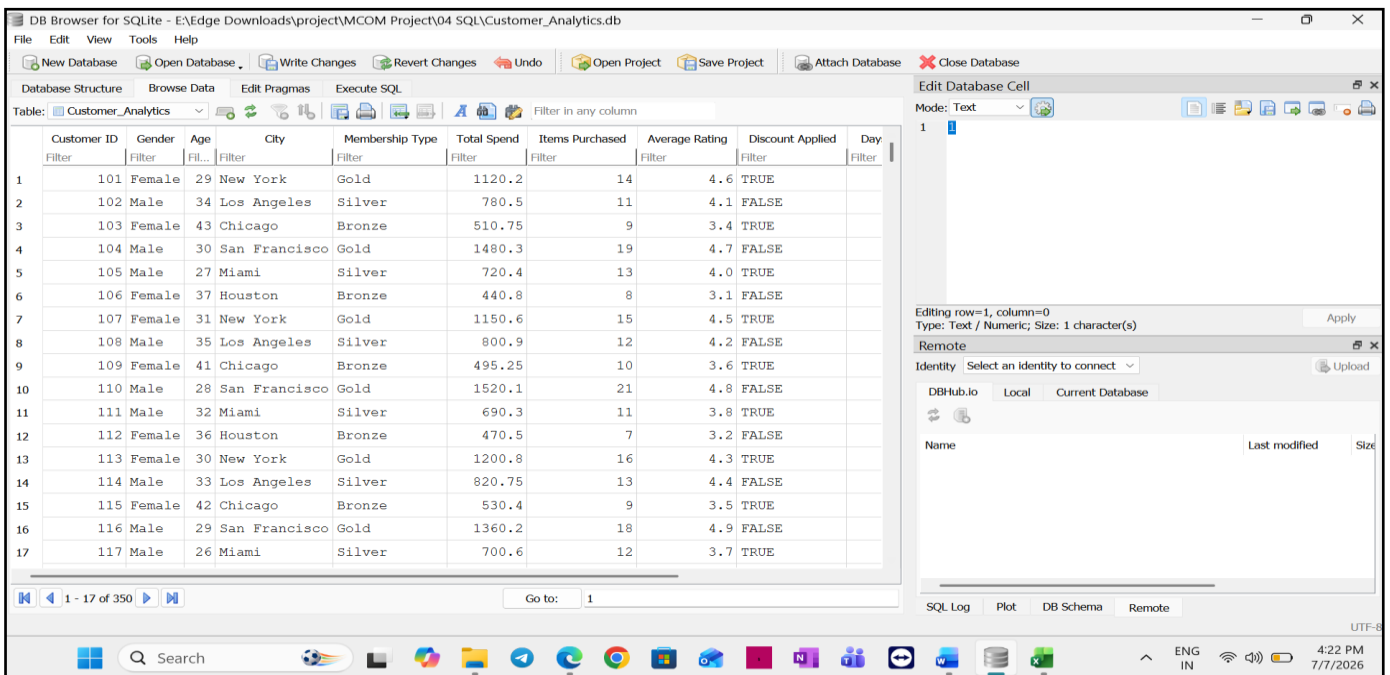
Overall, the relationship analysis reveals clear differences in customer spending and engagement across membership categories, geographical locations and promotional strategies. Gold membership customers consistently demonstrated higher spending behavior, while customer satisfaction remained relatively high irrespective of discount availability. These findings provide preliminary evidence of behavioral patterns that are further explored through SQL-based querying and interactive Power BI visualizations in the subsequent sections.

5.4 SQL-Based Business Analysis

SQL was used to perform structured querying and business-oriented analysis of the customer dataset. While Microsoft Excel was primarily used for exploratory analysis and summarization, SQL enabled efficient retrieval, aggregation and validation of customer information using query-based techniques. The analysis focused on identifying customer segments, spending behavior and purchasing patterns that support marketing decision-making in the online retail industry.

5.4.1 Data Exploration

The dataset was first explored using SQL to verify its structure and understand the distribution of records before conducting business-oriented analysis. Queries were



The screenshot displays the DB Browser for SQLite interface. The main window shows a table named 'Customer_Analytics' with 17 rows of data. The table has 10 columns: Customer ID, Gender, Age, City, Membership Type, Total Spend, Items Purchased, Average Rating, Discount Applied, and Day. The data is sorted by Customer ID. The right sidebar shows the 'Edit Database Cell' dialog box, which is currently empty. The bottom status bar indicates the current row is 1 of 17, and the total number of rows is 350.

	Customer ID	Gender	Age	City	Membership Type	Total Spend	Items Purchased	Average Rating	Discount Applied	Day
1	101	Female	29	New York	Gold	1120.2	14	4.6	TRUE	
2	102	Male	34	Los Angeles	Silver	780.5	11	4.1	FALSE	
3	103	Female	43	Chicago	Bronze	510.75	9	3.4	TRUE	
4	104	Male	30	San Francisco	Gold	1480.3	19	4.7	FALSE	
5	105	Male	27	Miami	Silver	720.4	13	4.0	TRUE	
6	106	Female	37	Houston	Bronze	440.8	8	3.1	FALSE	
7	107	Female	31	New York	Gold	1150.6	15	4.5	TRUE	
8	108	Male	35	Los Angeles	Silver	800.9	12	4.2	FALSE	
9	109	Female	41	Chicago	Bronze	495.25	10	3.6	TRUE	
10	110	Male	28	San Francisco	Gold	1520.1	21	4.8	FALSE	
11	111	Male	32	Miami	Silver	690.3	11	3.8	TRUE	
12	112	Female	36	Houston	Bronze	470.5	7	3.2	FALSE	
13	113	Female	30	New York	Gold	1200.8	16	4.3	TRUE	
14	114	Male	33	Los Angeles	Silver	820.75	13	4.4	FALSE	
15	115	Female	42	Chicago	Bronze	530.4	9	3.5	TRUE	
16	116	Male	29	San Francisco	Gold	1360.2	18	4.9	FALSE	
17	117	Male	26	Miami	Silver	700.6	12	3.7	TRUE	

Figure 5.11: Dataset Preview in DB Browser for SQLite

5.4.2 Descriptive SQL Analysis

SQL aggregate functions were used to summarize customer behavior and validate the descriptive findings obtained from Microsoft Excel. Functions such as **COUNT ()**, **AVG ()**, **MIN ()** and **MAX ()** were applied to analyze customer age, spending behavior, purchase quantity, customer ratings and purchase recency. The SQL outputs were consistent with the corresponding descriptive findings generated in Excel, providing cross-tool validation and supporting the consistency of the analytical results.

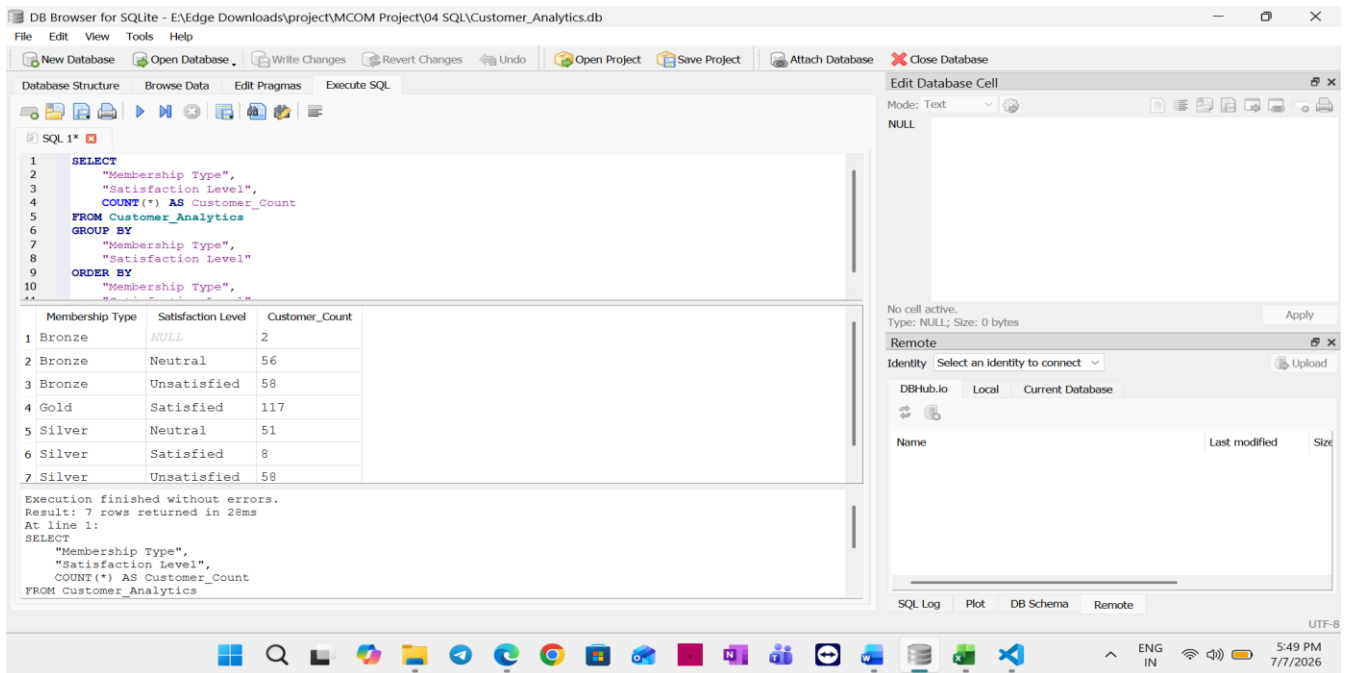


Figure 5.12: SQL Analysis of Customer Satisfaction by Membership Type

5.4.3 Business-Oriented SQL Queries

Beyond descriptive analysis, SQL was used to answer business-oriented questions that provide deeper insights into customer behavior. These analytical queries support marketing decision-making by identifying high-value customers, inactive customers and spending patterns across different customer segments.

5.4.3.1 Top Spending Customers

SQL queries were used to identify the highest-spending customers within the dataset. The analysis revealed that the leading customers recorded spending levels of approximately **1500.10**, indicating the presence of a small segment of highly valuable customers. Identifying such customers enables organizations to design premium loyalty programs and personalized retention strategies.

The screenshot displays the DB Browser for SQLite application. The main window shows a SQLite database named 'Customer_Analytics.db'. The 'Execute SQL' tab is active, displaying the following SQL query:

```
1 SELECT
2   "Customer ID",
3   City,
4   "Membership Type",
5   "Total Spend"
6 FROM Customer_Analytics
7 ORDER BY "Total Spend" DESC
8 LIMIT 10;
```

Below the query, the results are shown in a table with 4 columns: Customer ID, City, Membership Type, and Total Spend. The table contains 10 rows of data, all from San Francisco with Gold membership. The top two customers have a total spend of 1500.1.

Customer ID	City	Membership Type	Total Spend
188	San Francisco	Gold	1500.1
218	San Francisco	Gold	1500.1
260	San Francisco	Gold	1500.1
290	San Francisco	Gold	1500.1
331	San Francisco	Gold	1500.1
146	San Francisco	Gold	1490.1
176	San Francisco	Gold	1490.1

Below the table, the execution status is shown: 'Execution finished without errors. Result: 10 rows returned in 21ms. At line 1: SELECT "Customer ID", City, "Membership Type", "Total Spend"'. The right sidebar shows the 'Edit Database Cell' window with 'Mode: Text' and 'NULL' content. The bottom status bar shows 'ENG IN', '7/7/2026', and '6:08 PM'.

Figure 5.13 Top 10 Customers by Total Spend Using SQL

5.4.3.2 High-Value Customers Analysis

The analysis identified **117 customers** who recorded high spending despite not receiving promotional discounts. This finding suggests that customer purchasing decisions are influenced by factors beyond price reductions, including product quality, brand preference, convenience and overall customer experience. Such customers represent an important segment for long-term relationship management.

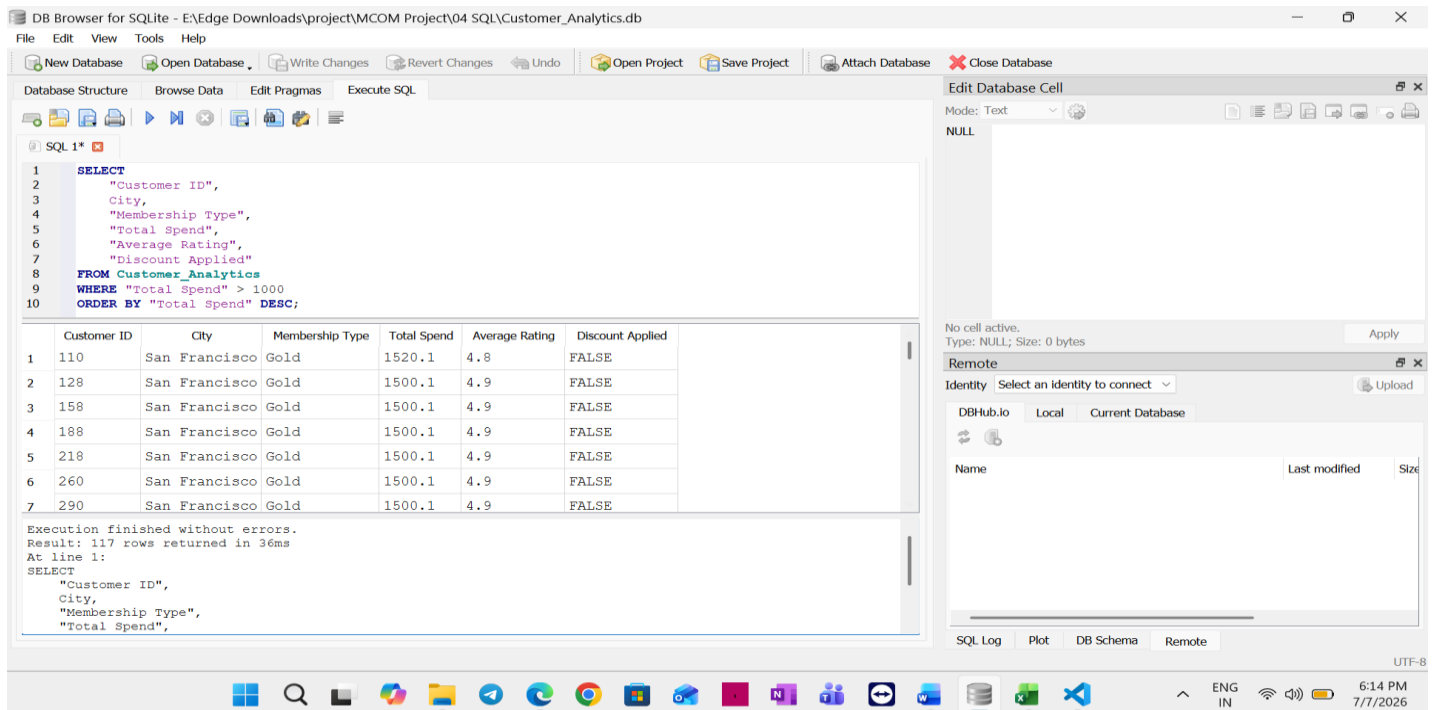


Figure 5.14 Customers with Total Spend > 1000

5.4.3.3 Inactive Customers

SQL analysis identified **124 customers** who had not made a purchase for more than **30 days**. These customers represent a potential re-engagement segment and may benefit from targeted promotional campaigns, personalized communication and loyalty initiatives aimed at encouraging repeat purchases.

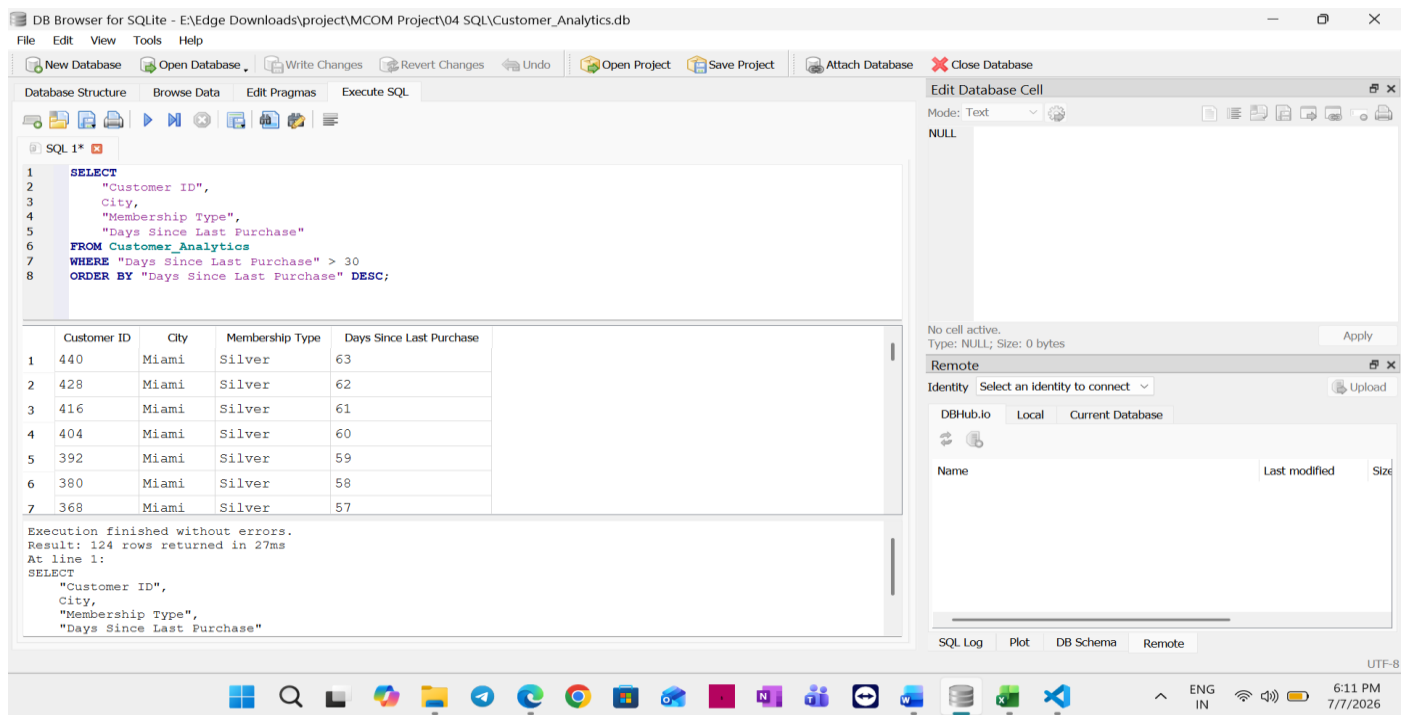


Figure 5.15 Customers with More Than 30 Days Since Last Purchase

5.4.4 Summary of SQL Findings

Overall, SQL provided an efficient platform for validating descriptive statistics and performing business-oriented analysis that would be difficult to conduct manually. The query-based approach enabled structured exploration of customer behavior, identification of valuable customer segments and extraction of actionable business insights. These findings complement the exploratory analysis conducted in

Microsoft Excel and provide the foundation for the interactive visualization presented through the Power BI dashboard.

The SQL scripts developed for data exploration, descriptive analysis and business-oriented analysis formed an integral part of the analytical workflow adopted in this study. The complete SQL source files constitute the supporting project documentation prepared by the researcher.

5.5 Power BI Dashboard Analysis

Microsoft Power BI was used to develop an interactive dashboard that integrates key customer metrics and visualizations into a single analytical interface. The dashboard combines descriptive measures, comparative charts and interactive filters to facilitate the interpretation of customer behavior and support data-driven marketing decisions. The visualizations developed in Power BI complement the analyzes performed in Microsoft Excel and SQL by presenting the findings in an interactive and decision-oriented format.

5.5.1 Key Performance Indicators (KPIs)

The dashboard includes Key Performance Indicator (KPI) cards summarizing important customer metrics, including total customers, average customer spending, average customer rating, average number of items purchased and average days since the last purchase. These KPIs provide a concise overview of customer behavior and

serve as the foundation for evaluating overall marketing performance and customer engagement. The indicators suggest that the customer base demonstrates relatively high satisfaction levels with moderate purchasing activity, while customer recency highlights opportunities for retention initiatives.

5.5.2 Average Customer Spend by Membership Type

The dashboard confirms that customer spending increases considerably with higher membership levels. Gold members recorded the highest average spending, followed by Silver and Bronze members. The visualization reinforces the effectiveness of customer loyalty programs in encouraging higher purchase values and suggests that premium membership categories are associated with higher observed customer spending and may therefore be relevant for customer-value-based segmentation.

5.5.3 Average Customer Spend by City

Geographic analysis demonstrates considerable variation in customer spending across different cities. San Francisco and New York recorded the highest average spending, whereas Houston and Chicago exhibited comparatively lower spending levels. These regional differences indicate that customer purchasing behavior varies across markets and may require location-specific marketing strategies and promotional campaigns.

5.5.4 Customer Satisfaction Distribution

The dashboard illustrates the distribution of customer satisfaction levels across the dataset. Satisfied customers represent the largest customer segment, although a substantial proportion of customers remain neutral or dissatisfied. This distribution indicates opportunities for organizations to strengthen customer relationships by improving service quality, product experience and post-purchase engagement.

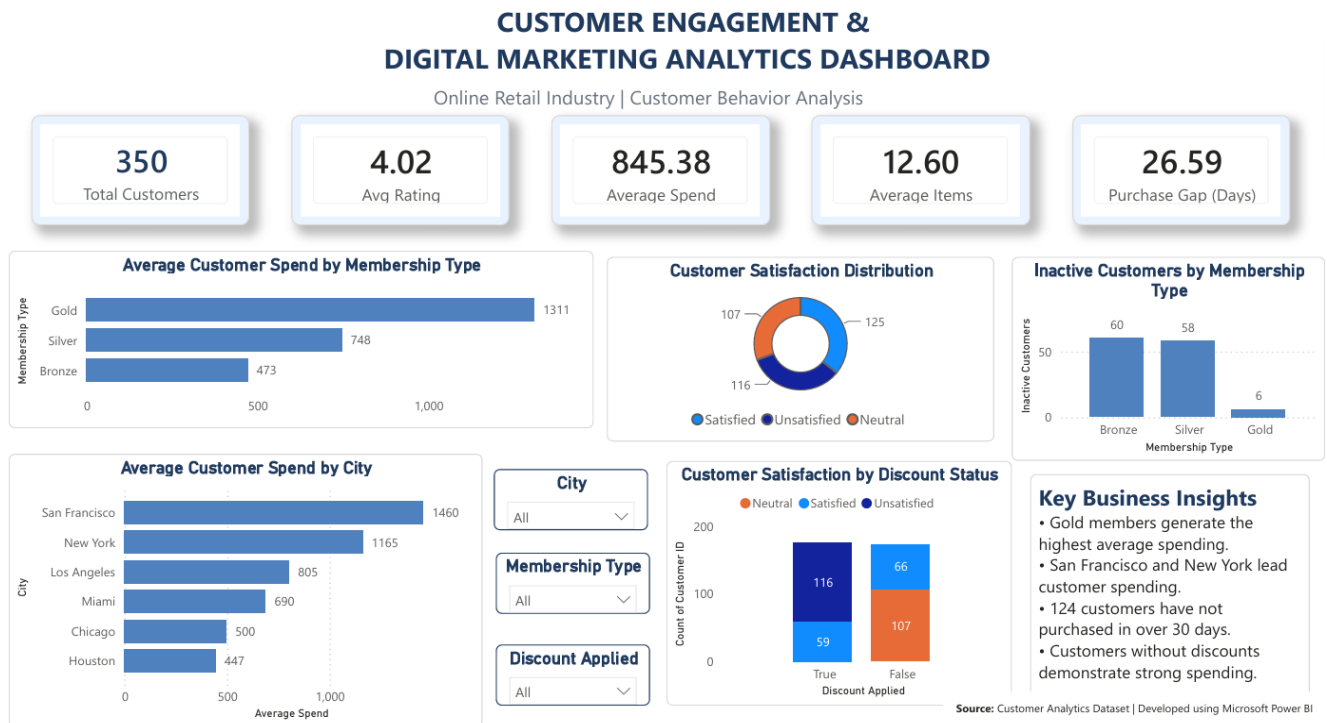
5.5.5 Customer Satisfaction by Discount Status

The comparison shows differences in satisfaction levels between customers who received discounts and those who did not. However, the observed pattern does not establish that discount application alone determines customer satisfaction. Other factors identified in the literature, such as product quality, service experience and trust, may also influence customer satisfaction.

5.5.6 Inactive Customers by Membership Type

The dashboard identifies customers who have not made a purchase for more than thirty days across different membership categories. This visualization enables organizations to identify customer segments requiring re-engagement strategies and supports targeted marketing initiatives designed to improve customer retention and repeat purchases.

Figure 5.16 Customer Engagement and Marketing Analytics Dashboard



5.6 Discussion of Findings

The exploratory analysis conducted using Microsoft Excel, SQL and Power BI demonstrate consistent customer behavior patterns across the dataset. Excel facilitated the initial exploration of customer characteristics and relationships, SQL enabled structured querying and validation of business-oriented insights, while Power BI integrated the findings into an interactive dashboard for managerial interpretation.

The analysis revealed that customer loyalty programs are associated with higher spending behavior, as Gold members consistently demonstrated greater average

expenditure than Silver and Bronze members. Geographical analysis further indicated considerable variation in spending patterns across cities, suggesting the need for region-specific marketing strategies.

Customer satisfaction remained relatively high throughout the dataset; however, the presence of neutral and dissatisfied customers indicates opportunities for improving customer experience and strengthening long-term engagement. Furthermore, SQL analysis identified inactive customers and high-value customers purchasing without discounts, highlighting opportunities for personalized retention strategies and targeted promotional campaigns.

Overall, the findings support the growing importance of business analytics in evaluating digital marketing performance and understanding customer engagement. The integration of spreadsheet analysis, query-based exploration and interactive visualization demonstrates how analytical tools can support evidence-based managerial decision-making in the online retail industry.

5.7 Key Findings

- Gold membership customers recorded the highest average customer spending among all membership categories.

- Customer spending varied significantly across different cities, with San Francisco and New York exhibiting the highest average expenditure.
- Overall customer satisfaction remained relatively high, although a substantial proportion of customers reported neutral or dissatisfied experiences.
- Promotional discounts influenced customer satisfaction but were not the sole determinant of positive customer experience.
- SQL analysis identified 124 inactive customers, indicating opportunities for targeted customer retention initiatives.
- A total of 117 customers recorded total spending above 1000, identifying a high-value customer segment that may be considered for differentiated retention and loyalty strategies.
- The integrated use of Microsoft Excel, SQL and Power BI enhanced the interpretation of customer behavior and supported data-driven marketing decision-making.

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 Summary of Findings

The present study examined digital marketing strategies and customer engagement in the online retail industry through a combination of conceptual review and exploratory analysis of a publicly available secondary dataset. The study adopted a descriptive and analytical research design and utilized Microsoft Excel, SQL and Microsoft Power BI to organize, analyze and visualize customer behavior and engagement patterns.

The review of literature highlighted that digital marketing has transformed the way organizations interact with customers by enabling personalized communication, multi-channel engagement and data-driven decision-making. Existing studies further emphasized that customer engagement is influenced by factors such as personalization, user experience, content quality, trust and effective communication strategies.

The exploratory analysis complemented these conceptual findings by providing practical insights into customer behavior. The analysis revealed that customer spending varied considerably across membership categories, with Gold members

recording the highest average expenditure, followed by Silver and Bronze members. Geographic analysis also indicated significant differences in customer spending across cities, with San Francisco and New York exhibiting comparatively higher average spending than other locations.

Customer satisfaction analysis suggested that although satisfied customers constituted the largest customer segment, a considerable proportion of customers reported neutral and dissatisfied experiences, indicating opportunities for strengthening customer engagement strategies. SQL-based analysis further identified inactive customers and high-value customers who made purchases without promotional discounts, demonstrating that customer loyalty extends beyond price-based incentives.

The Power BI dashboard integrated the analytical findings into an interactive visualization that supported the interpretation of customer behavior, spending patterns and engagement indicators. Overall, the study demonstrates that combining conceptual understanding with business analytics techniques can provide meaningful insights for evaluating marketing performance and supporting customer-centric decision-making in the online retail industry.

6.2 Conclusion

The study concludes, based on the reviewed literature, that digital marketing strategies play an important role in enhancing customer engagement within the online retail industry. The exploratory analysis conducted in the present study using Microsoft Excel, SQL and Microsoft Power BI further demonstrated that customer behavior and engagement-related indicators vary across membership categories, geographical locations, purchasing patterns and satisfaction levels.

Gold membership customers consistently exhibited higher spending behavior, while regional variations highlighted opportunities for location-specific marketing strategies. The identification of inactive customers and high-value customers purchasing without discounts further emphasized the importance of customer segmentation and targeted engagement initiatives.

The study also demonstrates that business analytics serves as an effective support mechanism for marketing decision-making by enabling organizations to identify behavioral patterns, monitor customer engagement and evaluate marketing performance using structured analytical techniques. The interactive dashboard developed in Power BI illustrates how analytical findings can be presented in a clear and accessible manner to support managerial decisions.

Overall, the study concludes that integrating digital marketing strategies with business analytics enables organizations to better understand customer behavior, strengthen customer engagement and support sustainable growth in the increasingly competitive online retail environment.

CHAPTER 7

RECOMMENDATIONS

7.1 Strategic Recommendations

Based on the conceptual review and the exploratory analysis conducted in the present study, the following strategic recommendations are suggested for improving digital marketing effectiveness and customer engagement in the online retail industry:

- Organizations should strengthen customer loyalty programs by designing personalized benefits and exclusive rewards for high-value customer segments to encourage repeat purchases and long-term retention.
- Digital marketing campaigns should incorporate customer segmentation based on spending behavior, membership category and geographical location to deliver more relevant and targeted promotional content.
- Businesses should continuously monitor customer satisfaction indicators and engagement metrics to identify areas requiring service improvement and customer experience enhancement.

- Organizations should utilize business analytics tools to evaluate customer behavior, monitor marketing performance and support evidence-based decision-making.
- Marketing strategies should adopt an integrated multi-channel approach that combines social media, search engine marketing, email marketing and personalized communication to provide a consistent customer experience across digital platforms.

7.2 Managerial Recommendations

- ❖ Managers should regularly analyze customer behavior using dashboards and analytical reports to support timely and informed business decisions.
- ❖ Customer segments identified as inactive should be targeted through personalized retention campaigns, promotional offers and customer re-engagement initiatives.
- ❖ High-value customers should receive differentiated marketing strategies, including exclusive loyalty programs, early access to new products and personalized communication to strengthen long-term customer relationships.
- ❖ Organizations should periodically evaluate regional differences in customer behavior and customize marketing strategies according to local market characteristics.

- ❖ Businesses should continue investing in analytical capabilities and digital technologies that enable real-time monitoring of customer engagement and marketing performance while ensuring responsible and ethical use of customer data.

CHAPTER 8

IMPLICATIONS OF THE STUDY

8.1 Academic Implications

The present study contributes to the academic understanding of digital marketing and customer engagement by integrating theoretical concepts with exploratory business analytics. It demonstrates how secondary data can be analyzed using analytical tools to support the interpretation of customer behavior within the online retail industry. The study also illustrates the practical application of business analytics techniques in commerce research, thereby encouraging greater integration of analytical approaches into marketing and management studies.

8.2 Practical Implications

From a practical perspective, the study highlights how organizations can utilize customer data to understand purchasing behavior, evaluate customer engagement and support marketing decision-making. The exploratory analysis identified differences in customer spending across membership categories and geographical locations, enabling businesses to design targeted marketing strategies and customer retention initiatives. The interactive Power BI dashboard further demonstrates how

business intelligence tools can assist managers in monitoring key performance indicators and communicating analytical findings effectively.

8.3 Broader Implications

The study reflects the increasing importance of digital transformation and data-driven decision-making in the online retail industry. As organizations continue to generate large volumes of customer data, business analytics is becoming an essential component of strategic marketing and customer relationship management. The findings of the study reinforce the need for organizations to combine digital marketing practices with analytical capabilities in order to enhance customer engagement, improve business performance and remain competitive in rapidly evolving digital markets.

CHAPTER 9

LIMITATIONS AND FUTURE SCOPE

9.1 Limitations of the Study

Although the study provides useful insights into digital marketing strategies and customer engagement in the online retail industry, certain limitations should be acknowledged.

- The study is primarily based on secondary data and published literature, which limits direct validation of customer perceptions through primary research.
- The exploratory analysis was conducted using a publicly available e-commerce customer behavior dataset. Therefore, the findings are illustrative in nature and may not represent the characteristics of all online retail organizations or customer populations.
- The study adopted descriptive and exploratory analytical techniques using Microsoft Excel, SQL and Microsoft Power BI. Advanced statistical methods such as regression analysis, predictive modelling or machine learning techniques were beyond the scope of the present study.

- The analysis focused on selected customer variables available in the dataset. Other factors influencing customer engagement, such as income level, product category, browsing behavior and purchase frequency, were not available for analysis.
- The findings should therefore be interpreted within the context of the selected dataset and the objectives of the present research.

9.2 Scope for Future Research

Future research may extend the present study in several ways:

- Primary data may be collected through surveys or interviews to validate and compare the findings obtained from secondary data.
- Researchers may apply advanced statistical techniques, predictive analytics or machine learning models to improve customer segmentation and behavioral prediction.
- Future studies may analyze larger and more diverse datasets obtained from multiple online retail platforms to improve the generalizability of findings.
- Additional customer variables such as purchasing frequency, browsing behavior, demographic characteristics and product preferences may be

incorporated to develop a more comprehensive understanding of customer engagement.

- Comparative studies across different industries or geographical regions may further enhance understanding of digital marketing strategies and customer behavior in diverse business environments.

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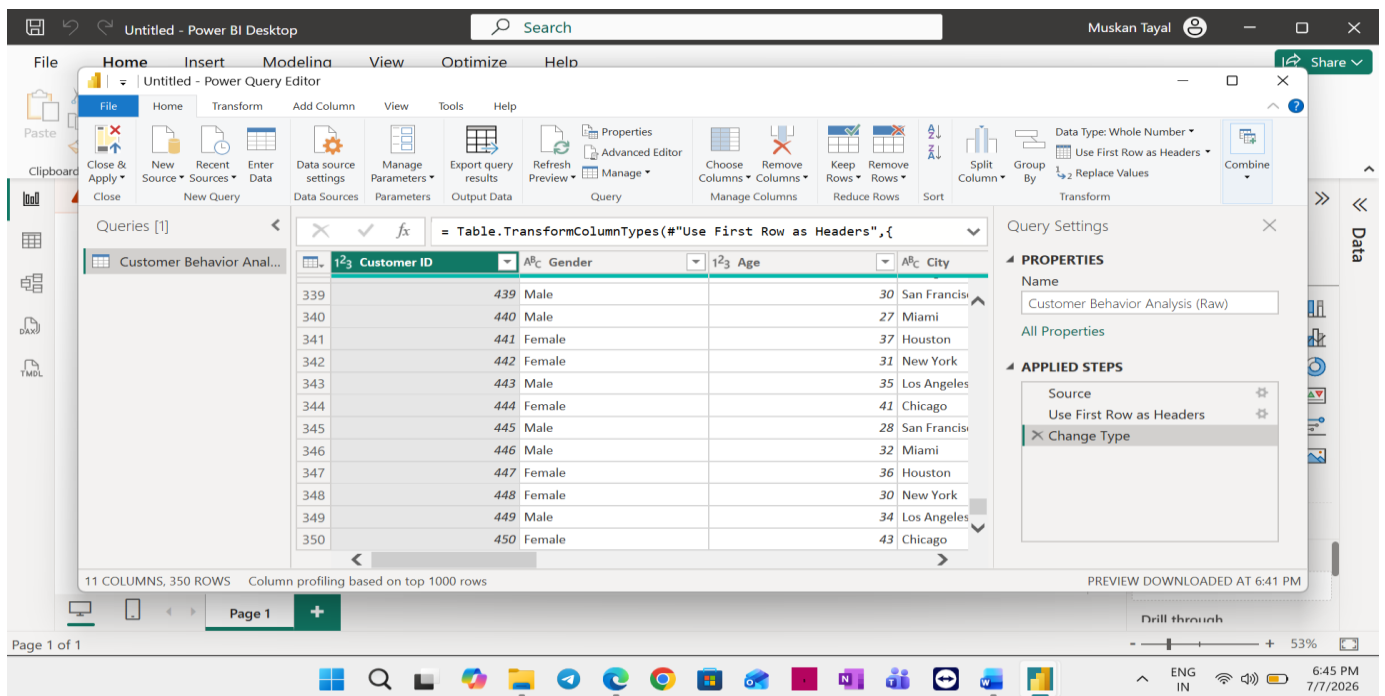
Kaggle. <https://www.kaggle.com/datasets/uom190346a/e-commerce-customer-behavior-dataset>

APPENDICES

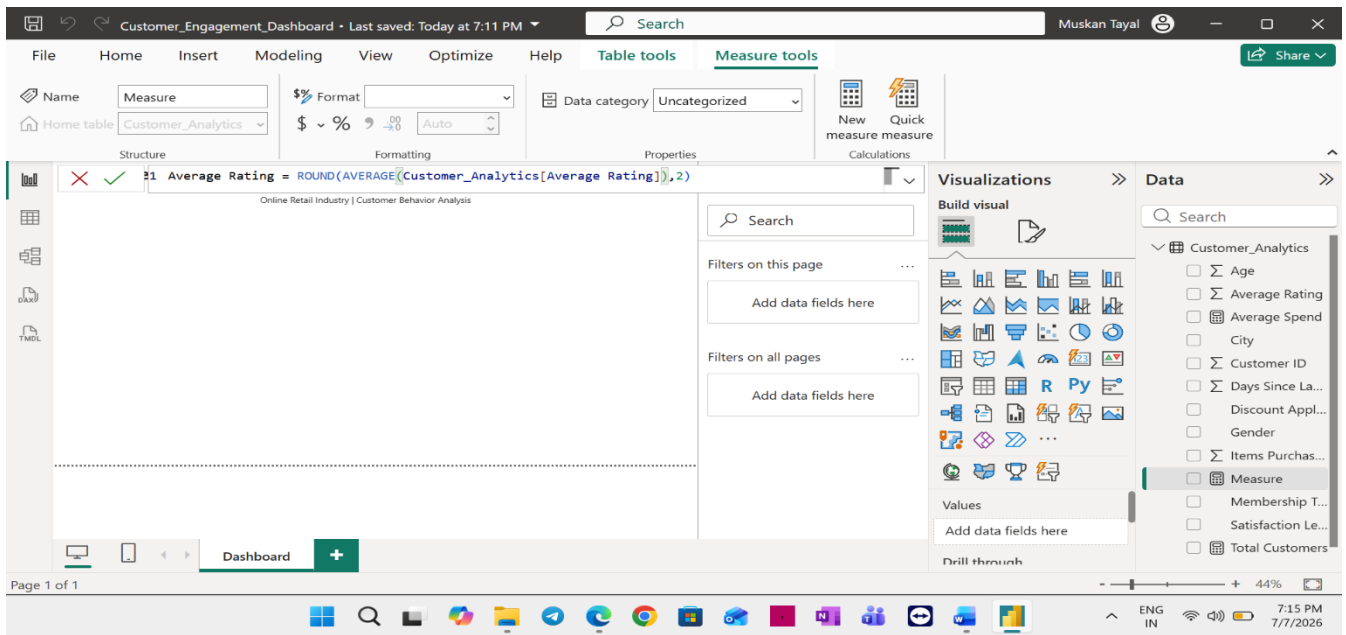
Appendix A: Project Files

The analytical workflow adopted in the present study involved the use of Microsoft Excel, SQL (DB Browser for SQLite) and Microsoft Power BI. The following files were developed as part of the project:

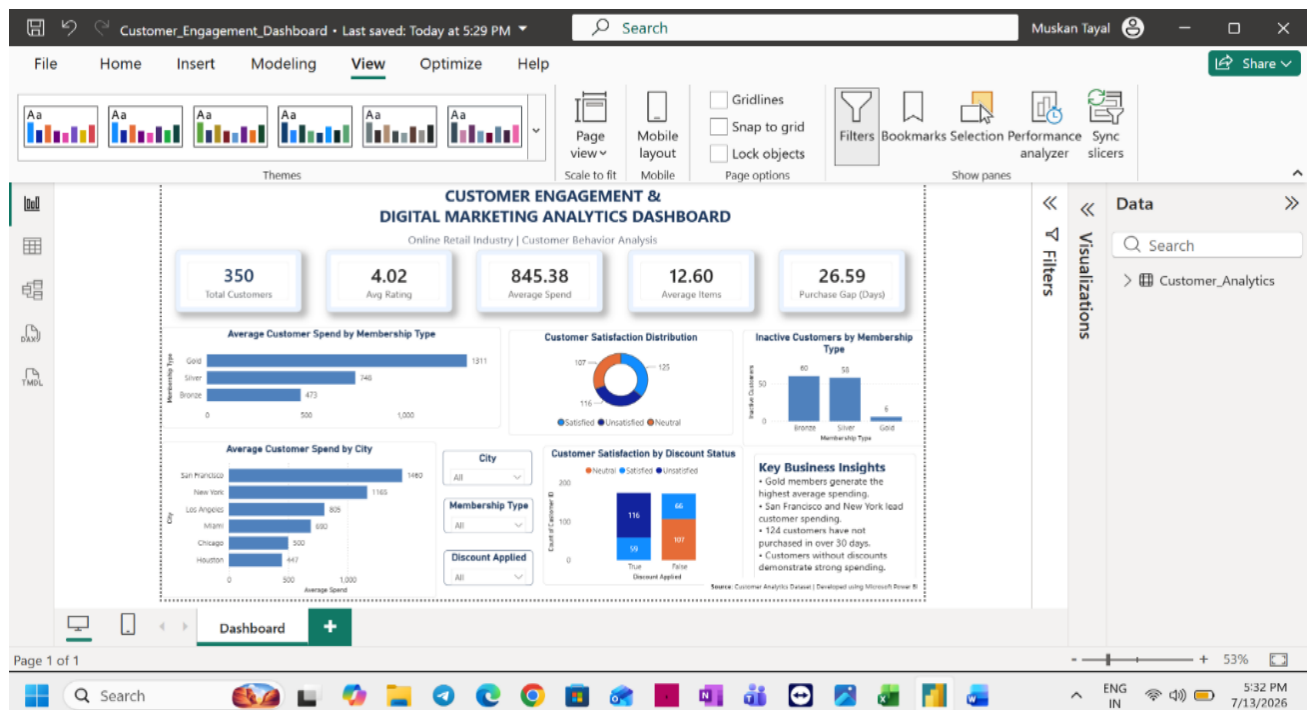
- **Microsoft Excel workbook** containing data preparation, descriptive statistics and PivotTable-based exploratory analysis.
- **SQL source files** containing data exploration, descriptive analysis and business-oriented queries.
- **Microsoft Power BI dashboard (.pbix)** developed for interactive visualization and managerial interpretation of customer behavior.



Appendix Figure A.1 Power Query Data Preparation



Appendix Figure A.2 Power BI Measures



Appendix Figure A.3 Power BI Dashboard Development Environment